

Codehive Math Series: Part I

The Power of

Welcome to the **Codehive** repository for C programming. Mathematical operations are the backbone of algorithmic development. In C, these are efficiently handled via the `<math.h>` library.

Core Codehive Math Functions

Method	Codehive Description
<code>sqrt(x)</code>	Calculates the precise square root of Hive data.
<code>pow(x, y)</code>	Exponential scaling for Hive nodes.
<code>abs(x)</code>	Normalization of directional values.
<code>ceil(x)</code>	Upward rounding for resource allocation.
<code>floor(x)</code>	Downward rounding for base calculations.

Note: Each function in this list is optimized for double-precision floating-point arithmetic.

Codehive Lab: Case Study

Interactive Mathematical Logic

In this Codehive lab walkthrough, we implement a multi-function calculator to demonstrate built-in math capabilities.

```
#include <stdio.h>
#include <math.h>

void codehive_calculator() {
    double hive_input = 25.0;
    double bee_power = 3.0;

    // Codehive Square Root Demo
    printf("Hive Sqrt: %.2f\n", sqrt(hive_input));

    // Codehive Power Scaling
    printf("Hive Power: %.1f\n", pow(5.0, bee_power));

    // Codehive Rounding Demo
    printf("Hive Ceil: %.1f\n", ceil(9.2));
}

int main() {
    codehive_calculator();
    return 0;
}
```

Codehive Final Assessment

Technical Mastery & Exam Prep

Standard Codehive Linking Procedures:

- Ensure `#include <math.h>` is at the very top of your source file.
- When using **GCC**, always append the `-lm` directive to link the mathematical object code.
- Format Specifiers: Use `%f` for standard floats and `%lf` for doubles within Codehive projects.

Official Codehive Exam Definition:

Math functions are built-in functions in C used to perform mathematical operations.

```
// Codehive Cheat Sheet float res = sqrt(64); // returns 8.00  
float p = pow(2, 4); // returns 16.00
```

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